

F.R.I.D.A.Y

Flying Relay Infrastructure for Dynamic Area Coverage
Drone-Based Aerial Relay System for Disaster Response Communication

1. Idea Title, Core Problem Statement & Innovation Summary

1.1 Idea Title

F.R.I.D.A.Y. — Flying Relay Infrastructure for Disaster Area sYstem

Drone-Based Aerial Relay System for Disaster Response Communication.

1.2 Core Problem Statement

During a disaster one of the most vulnerable systems is the communication infrastructure. Cyclones, floods, earthquakes and landslides may damage towers and power failures can disable base stations and leave affected regions with lack of connectivity for several hours or even days. When this happens rescue operations face delays, coordination issues and people who are located in inaccessible areas may not be able to communicate their location or condition. Similarly last-mile connectivity limitations are also encountered in dense forest areas, hilly regions and large agricultural zones where installing and maintaining permanent communication infrastructure would be economically impractical. Currently existing solutions such as satellite phones, fixed repeater stations and conventional short-range radio systems face shortcomings like high deployment costs, longer setup times and dependence on infrastructure that may itself be damaged or unavailable during the emergency.

1.3 Innovation Summary

F.R.I.D.A.Y. is designed as an aerial communication relay in which a drone is used to elevate a communication node above the ground, effectively functioning as a mobile and rapidly deployable communication tower. Two independent long-range radio modules are carried by the drone, with one providing a communication link to a forward ground station and the other establishing a link with the command centre. Through this configuration, two otherwise disconnected locations can be bridged using a single aerial relay node.

The system is designed to operate within a license-free radio band and does not require pre-existing network infrastructure for its operation. Two-way communication is supported, while signal-strength measurements are utilised to provide an estimation of the distance between connected nodes. System status and communication parameters can be monitored through a local web-based dashboard, which can be accessed using a smartphone or laptop without requiring an internet connection. The proposed architecture also provides scope for future integration of encrypted messaging, multi-drone mesh networking, and existing disaster-alert systems.

2. Proposed Technical Solution & Concept Notes

This section describes the system architecture, working principle and engineering approach along with technical details.

2.1 System Architecture

The system is made of three nodes operating in a 868 MHz licence-free ISM band using LoRa (Long Range) radio modulation:

- Ground Station A: transmits status or distress data using directional antenna
- Aerial Relay Node (Drone): carries two radio modules on a single microcontroller: one dedicated to receiving from Ground Station A, the other dedicated to forwarding data to Ground Station C. Each radio operates on an isolated channel to prevent cross-interference on shared hardware. An LPDA is mounted on the drone relay to increase
- Ground Station C: receives relayed data and can transmit replies back through the same relay making the link fully bidirectional

Operation above ground-level using a drone relay ensures that obstructions such as terrain, forest canopy, damaged structures do not hinder the effective communication range and line-of-sight reach of waves is enhanced, directly addressing the 'last-mile' problem in disaster communication.

2.2 Key Technical Features

- Two-way communication: Both nodes can transmit and receive through the relay.
- Distance estimation based on RSSI: Live RSSI data is converted into an approximate range estimate at every hop without requiring GPS hardware.
- Local servers: Each node has its own WiFi access point and live web interface and from any phone or laptop it can be accessed with zero installation requirements and has no dependency on cellular or internet connectivity.
- One-tap emergency presets: For rapid distress signalling pre-made message templates of SOS, medical emergency and all-clear messages are available.
- Live delivery and signal-quality logging: Every hop's success/failure status, signal strength and estimated distance is logged and visible in real time.

2.3 Security and Reliability Roadmap

Currently the prototype transmits data over an open radio channel. The next planned enhancement is AES-128 message encryption using the microcontroller's built-in hardware cryptographic acceleration, paired with a lightweight integrity check to detect any tampering during transit to ensure that even if the radio signal is intercepted the content remains unreadable and any modification is detectable.

2.4 Scalability Roadmap (Mesh Extension)

The present system consists of a single aerial relay which connects two ground stations. The architecture is designed to scale into a multi-node mesh in which several relay units hop data between each other. This provides an extended coverage across a larger disaster-zone without requiring every station to be in direct range of every other station.

2.5 Disaster-Trigger Integration Roadmap

Human intervention and human reaction time can be reduced by integrating the current system with existing national early-warning infrastructure - India's Common Alerting Protocol (CAP) based public alert system. A verified disaster alert can automatically notify a field operator to deploy the relay. This reduces the detection-to-deployment window from hours to minutes.

2.6 Concept Diagrams

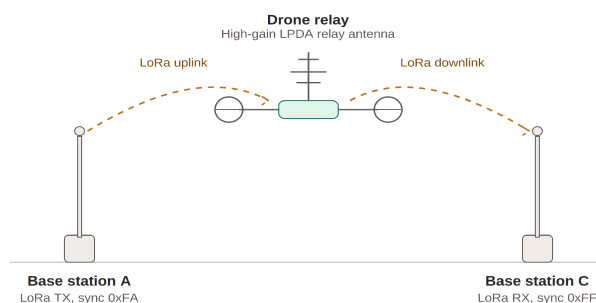


Fig. 1 — Conceptual layout: two ground stations bridged by an aerial relay node.

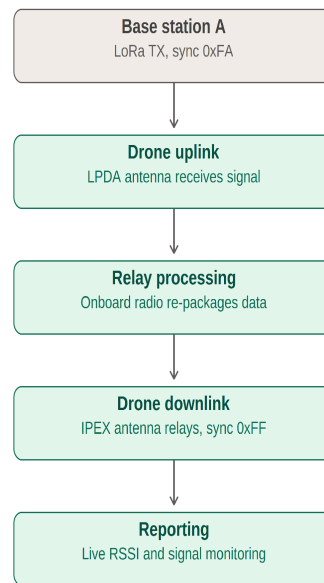


Fig. 2 — System architecture / data flow across the operational stages.

3. Pitch Deck Summary & Short Business Plan

3.1 Value Proposition

A rapidly deployable , license-free, infrastructure-independent aerial communication bridge, positioned as a low-cost middle ground between expensive satellite communication and slow-to-restore fixed tower infrastructure — deployable in minutes rather than days, with zero recurring spectrum cost.

3.2 Target Market Segments

- Disaster response agencies (state and national disaster response teams - SDRF and NDRF) requiring rapid, infrastructure-free coordination during relief operations.
- Defence and border security forces which require tether-independent communication in remote terrain, as a secondary application of the same core architecture.

3.3 Revenue Model

- Direct unit sales to government and private agencies.
- Per-deployment service fees

- Annual Maintenance Contract (AMC) based support and operator-training contracts for long-term partnerships.

3.4 Go-to Market Channels

Government tenders, defence procurement processes, direct agency outreach, and participation in innovation/incubation platforms which provide direct visibility with evaluators and early adopters in this domain.

4. Estimated Budget, Development Plan, Timeline & Expected Outcomes

4.1 Prototype Build Cost

Component	Quantity	Est.Cost (INR)
LPDA antenna	1	₹10,000
ESP32 microcontroller boards	3	₹1,200 – ₹1,800
SX1262 LoRa radio modules	4	₹4,000 – ₹6,000
IPEX omnidirectional antennas	3	₹300 – ₹600
Voltage regulation, capacitors, wiring, misc.	–	₹500 – ₹800
Total Prototype Build Cost		₹15,000 – ₹20,000

4.2 Field-Deployment Investment Breakdown (₹3,00,000 – ₹5,00,000)

Category	Est. Allocation (INR)
Production-grade RF component sourcing- LPDA, LoRa modules, matched gain states)	₹80,000 – ₹1,20,000
Drone airframe, motors, battery system and payload mounting suitable for rugged terrain	₹1,20,000 – ₹1,80,000
Weatherproof enclosures for ground stations and relay payload	₹25,000 – ₹40,000
Field range and reliability testing (multiple site trials)	₹30,000 – ₹50,000

Certification, compliance, and documentation	₹20,000 – ₹40,000
Contingency and iteration buffer	₹25,000 – ₹70,000
Total	₹3,00,000 – ₹5,00,000

4.3 Ongoing Operating Cost

Item	Est. Cost (INR)
Monthly operating expense(testing, maintenance, upkeep)	₹10,000 – ₹20,000
Projected breakeven period	~12 months from launch
Projected post-breakeven ROI multiple	1.2x-1.4x

4.4 Development Plan & Timeline

Phase	Milestone
Currently completed	Dual-radio aerial relay validated; two-way communication with RSSI-based distance estimation; local WiFi dashboards operational and mobile-accessible; emergency preset messaging implemented.
Mid-term (0-9 months)	Ruggedised hardware enclosure; extended field trials; GPS-based precise location tagging.
Long-term (9-18 months)	Multi-node mesh extension; integration with national disaster-alert (CAP-based) trigger systems; production-scale deployment.

4.5 Expected Outcomes

- Restores command-and-control communication in disaster-hit, network-dark zones.
- Provides a low-cost alternative to satellite and fixed-tower infrastructure for relief operations.
- Enables agricultural monitoring and rural connectivity without cellular dependency.

- The same core relay architecture is also applicable to defence and border-security field communication in remote terrain.

5. Prototype Status & Technical Demonstration

A working three-node hardware prototype has been built and validated on the bench: two ground/field stations and one dual-radio aerial relay node, communicating over the 868 MHz LoRa band. The system has been tested for bidirectional message relay, live RSSI-based distance estimation, and mobile-accessible local dashboard operation across all three nodes.

5.1 Prototype Images

